

Mohamed Amine Haouas

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Education

Software Engineering Degree

National Institute of Applied Science and Technology

Tunis, Tunisia

Sept 2022 – Fall 2027

Experience

Core Maintainer / Google Summer of Code 2026 OICR, Canada

May 2026 – Present

- Built **Camille**, a patient-facing agent that turns a plain-language cancer description into a ranked **clinical trial** search, streaming cited answers over **SSE** with every cited site plotted on a map.
- Engineered the async **FastAPI** + **Pydantic AI** backend on a layered architecture: filter-then-rank search over **PostgreSQL** + **pgvector**, **Redis** sessions, and a **React** + **TypeScript** SPA sharing one selection across its three panels.
- Hardened the system with a prompt-triage gateway, output guardrails and red-team scenarios using **Promptfoo**, and measured it with **Langfuse** tracing over versioned prompts plus a **DeepEval** evaluation harness.

Software Engineer Intern / Working Student Ubotica Technologies, Ireland

June 2026 – Present

- Built a **project-agnostic agent template** connecting agents to company projects over **MCP** and REST APIs. Used **Pydantic-AI** capabilities to load and manage tools and MCPs on demand per agent scope and task.
- Shipped a **natural-language-to-search-query agent** on it, integrated end to end into an internal **MLOps platform** for comparing training experiments.
- Developed the platform's data ingestion pipeline: chunking, validation, and third-party integrations for external sources.
- Helped refactor the training stack into a standalone containerized submodule run serverlessly via a **dispatcher-reconciler** architecture: **Vertex AI custom training jobs** for training, **Cloud Run** for playground inference.

Software Engineer Intern / Working Student Vectors (Atlassian Marketplace Partner), Tunisia

June 2025 – May 2026

- Cut **LLM workflow** latency by **50%** and led a provider-agnostic refactor with **Pydantic-AI** to benchmark and swap models seamlessly, instrumented with **Langfuse** for prompt evaluation and usage analytics.
- Built an internal **test-generation agent** that opens unit-test PRs on top of feature PRs for frontend components, enforcing **code coverage** and **type validation**; runs non-interactively with scoped skills and protection hooks.
- Led a small team to build **Vexel**, a design-sync product that aligns code components with intended designs and automatically generates and maintains up-to-date documentation.

Projects

Open Source Contribution – Pydantic-AI

[Github Repository](#)

- Contributed [3 merged PRs](#) to **Pydantic-AI**, an open-source LLM framework, adding run-time `builtin_tools` support.

Prospera: Financial Document Analysis Platform

[Github Repository](#)

- Built a full-stack platform (**Next.js**, **FastAPI**, **Celery**) that scrapes financial prospectuses and extracts structured data via a multi-stage pipeline with financial statement detection using **Azure Document Intelligence**.
- Developed a **RAG-based conversational search** system using **pgvector** for semantic retrieval.

Animoji: AI-Powered Anime Art Generation Platform

[Github Repository](#)

- Built a full-stack platform with a **React** + **shadcn/ui** SPA and a **Go** API gateway orchestrating uploads, auth, and real-time job streaming, deployed on **Azure** with **Ansible**-provisioned infrastructure.
- Designed an AI pipeline using **Azure OpenAI** for structured prompt enhancement feeding into **FLUX.2-pro** image synthesis, decoupled from the gateway via a **NATS** message bus and a **Python** worker.

Volunteering

Technical Project Manager – IEEE INSAT CS Chapter

[Github Organization](#)

- Led 6 engineering students in building **VICoach**, an AI-assisted platform for mock interviews to evaluate performance.

Honors and Awards

- **2nd Place** – IEEE TSYP 12 CS Technical Challenge , organized by **IEEE Tunisian Section**.
- **2nd Place** – AINS AI Hackathon, organized by **IEEE CS**.

Technologies

Languages: Python, JavaScript, TypeScript, Golang

Tools/Frameworks: Git, FastAPI, React, Docker, Vite, Pydantic-AI, MCP, DeepEval, PostgreSQL, pgvector, Redis, Sentry, Langfuse, OpenTelemetry, Nginx, Ansible, Azure, GCP